

15.2 Directed Graph



The core skills you'll master in this section:

- ▶ A directed graph is a structure made up of nodes and directed edges.
- ▶ A directed edge has direction — it represents a one-way relationship "pointing" from one node to another.
- ▶ The adjacency matrix of a directed graph is a square matrix, whose element in row i , column j represents the number (or weight) of edges pointing from node i to node j .
- ▶ The element in row i , column j of the k -th power of a directed graph's adjacency matrix represents the number of paths from node i to node j that pass through exactly k edges.
- ▶ In a directed graph, out-degree is the number of edges leaving a node, corresponding to the sum of each row of the adjacency matrix; in-degree is the number of edges pointing into a node, corresponding to the sum of each column.

This section covers directed graphs. Its structure closely parallels the previous section on undirected graphs, so we recommend reading the two sections side by side.

Directed Graph

In graph theory, a directed graph is a data structure made up of nodes and directed edges.

As mentioned in the previous section, an undirected graph's edges represent connections between nodes; a directed graph's directed edges, by contrast, represent a one-way relationship of "from where to where."

Take, for example, Figure 1 — suppose we have 4 cities, each of which can be regarded as a node. In Figure 1, this graph has four nodes in total: a , b , c , and d .

If a flight exists between two cities, a directed line segment is drawn connecting them, and that line segment is a directed edge.

For example, if there's a flight from a to b , we draw an arrow from a to b in the graph. At the same time, if there's also a flight from b to a , we draw another arrow from b to a .

Looking at Figure 1, we find that round-trip flights only exist between cities a and b , and between c and d .

Some cities have no direct flights between them. For example, flying from c to a is inconvenient — it requires the route $c \rightarrow d \rightarrow a$.

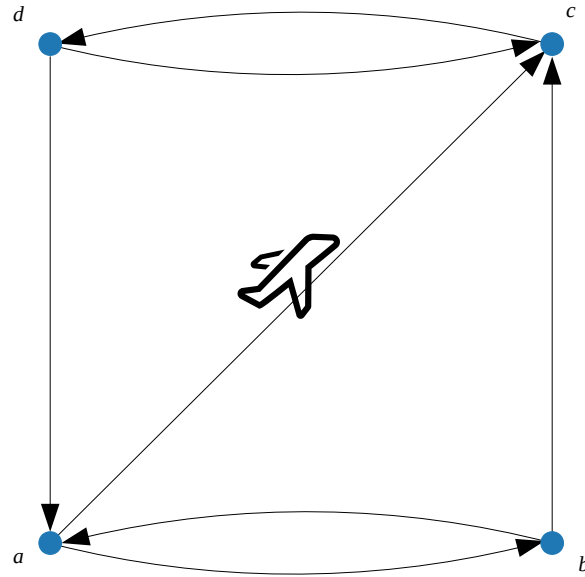


Figure 1. The Directed Graph Formed by Flights Between 4 Cities

Drawing a Directed Graph with NetworkX

Code 1 uses NetworkX to draw Figure 1 — this directed graph. Let's walk through the key statements below.

- a** Use `nx.DiGraph()` to create a directed graph object, named `directed_G`. In a directed graph, edges have direction — for example, the edge from `a` to `b` is a different (directed) edge from the one from `b` to `a`; the two directions are treated separately. `DiGraph` is short for "Directed Graph."
- b** Use `add_nodes_from()` to add four nodes to the graph at once, named `'a'`, `'b'`, `'c'`, and `'d'`. These names can be anything; they represent the elements in our graph — in this case, cities.
- c** Add directed edges to the graph. We add seven edges at once; each edge is an "arrow" from one node to another. For example, `('a','b')` represents an edge from node `'a'` to `'b'`. Note that both `('a','b')` and `('b','a')` appear here, meaning the two nodes are connected in both directions — there are flights or relationships going both ways. `add_edges_from` is a function for adding multiple edges at once.
- d** Define where each node should be displayed on the graph. We manually assign 2D coordinates to the four nodes `'a'`, `'b'`, `'c'`, and `'d'`.
- e** Used to visualize this directed graph. Use `draw_networkx` to draw the graph `directed_G`.
`pos=pos` means we use the node coordinates we just specified.
`node_size=188` sets the size of each node — the larger the number, the bigger the circle.
`connectionstyle='arc3, rad = 0.1'` is a small trick used to draw bidirectional edges as curved arcs instead of two overlapping straight lines. For example, when both `'a'` to `'b'` and `'b'` to `'a'` exist, this parameter makes them curve so they're both visible instead of overlapping.

Code 1. Drawing a Directed Graph with NetworkX |  LA_15_02_01.ipynb

```

## 初始化
import matplotlib.pyplot as plt
import networkx as nx

## 有向图
a directed_G = nx.DiGraph()

## 节点
b directed_G.add_nodes_from(['a', 'b', 'c', 'd'])

## 有向边
c directed_G.add_edges_from([('a', 'b'),
                             ('b', 'a'),
                             ('a', 'c'),
                             ('d', 'a'),
                             ('b', 'c'),
                             ('c', 'd'),
                             ('d', 'c')])

## 定义节点坐标
d pos = {'a': [0, 0],
         'b': [1, 0],
         'c': [1, 1],
         'd': [0, 1]}

## 可视化
e plt.figure(figsize = (6,6))
nx.draw_networkx(directed_G, pos = pos, node_size = 188,
                  connectionstyle = 'arc3, rad = 0.1')

```

Adjacency Matrix

A directed graph like Figure 1 can also be represented using an adjacency matrix.

Figure 1's adjacency matrix, as a directed graph, is

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

In simple terms, the adjacency matrix of a directed graph is a square matrix, whose element in row i , column j represents the number (or weight) of edges pointing from node i to node j (for a weighted directed graph). As shown in Figure 1 (a), an edge exists pointing from node a to b , so the element in row 1, column 2 of adjacency matrix A is 1.

⚠ Note the order: row first, then column.

As introduced in the previous section, the adjacency matrix of an undirected graph is always symmetric; a directed graph's adjacency matrix, by contrast, is generally not symmetric.

For a directed graph, its adjacency matrix is symmetric if and only if the reverse of every directed edge also exists in the graph. In other words, for any pair of nodes, if an edge exists from node a to node b ,

an edge must also exist from node b to node a — only then does the whole graph correspond to a symmetric adjacency matrix. In that case, although the graph is technically directed, it's structurally no different from an undirected graph.

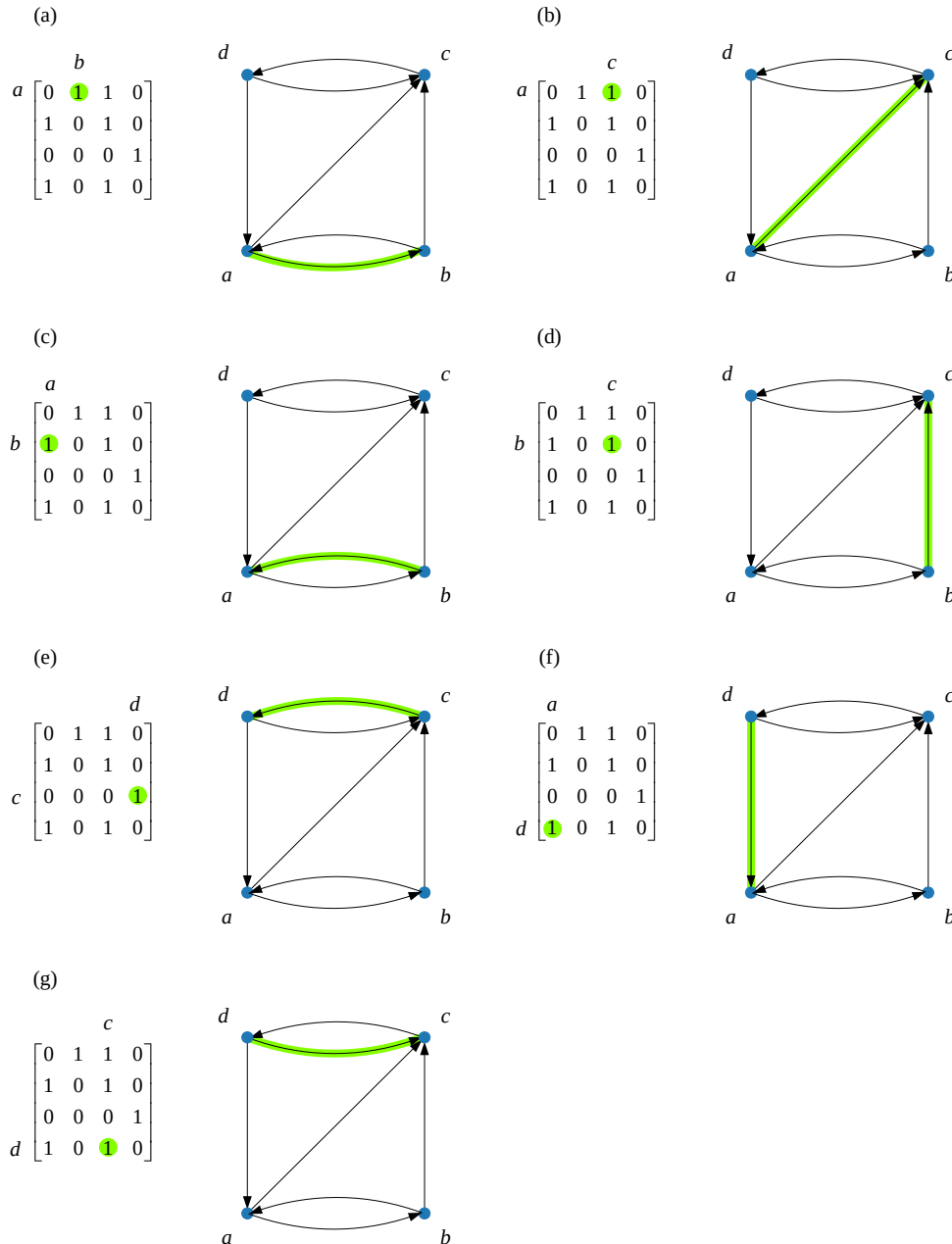


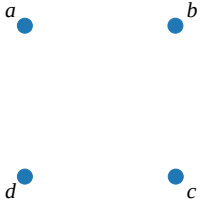
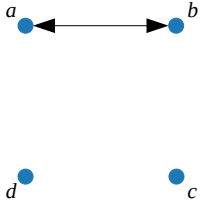
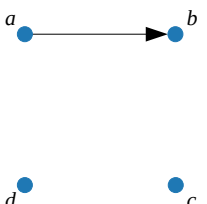
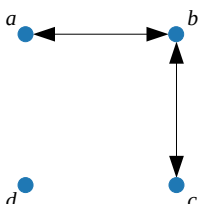
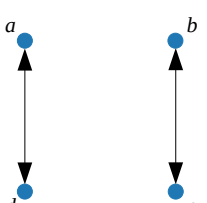
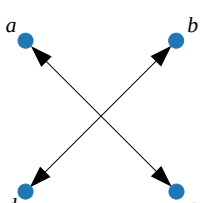
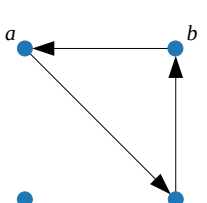
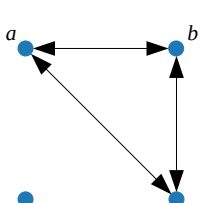
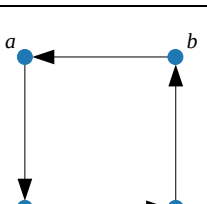
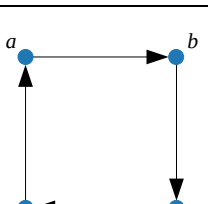
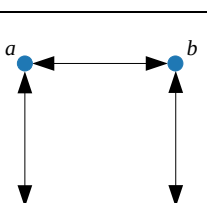
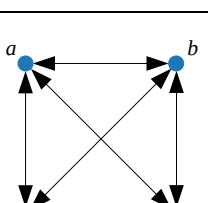
Figure 2. Each Element of the Directed Graph's Adjacency Matrix



LA_15_02_02.ipynb constructs a directed graph from an adjacency matrix (a 2D array); please study it on your own.

? Please analyze each directed graph and adjacency matrix in Table 1 one by one.

Table 1. Several Directed Graphs and Adjacency Matrices Constructed From 4 Nodes

Directed Graph	Adjacency Matrix	Directed Graph	Adjacency Matrix
	$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
	$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$		$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$
	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
	$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$		$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$
	$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$		$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

Looking Again at the Adjacency Matrix

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As shown in Figure 3, in row 1 of adjacency matrix $\mathbf{A}$, the elements equal to 1 represent the nodes that a can reach directly, one-way. In other words, without passing through any other node, node a can walk directly to b and c . This can be written as the following matrix multiplication

$$\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} @ \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} @ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$$

The equation above is equivalent to extracting the first row of matrix $\mathbf{A}$.

? Please analyze the remaining rows of adjacency matrix $\mathbf{A}$.

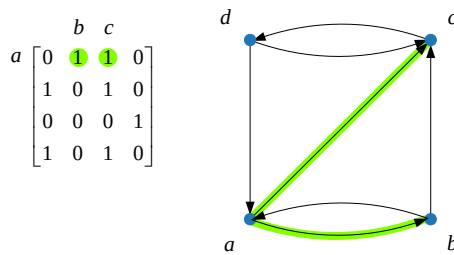


Figure 3. a Directly Reaches Two Nodes — Row 1 of the Directed Graph's Adjacency Matrix $\mathbf{A}$

As shown in Figure 4, column 1 of adjacency matrix $\mathbf{A}$ shows the two nodes that can directly reach a , corresponding to the following multiplication

$$\mathbf{A} @ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

The equation above is equivalent to extracting the first column of matrix $\mathbf{A}$.

? Please use the same approach to analyze the remaining columns of adjacency matrix $\mathbf{A}$.

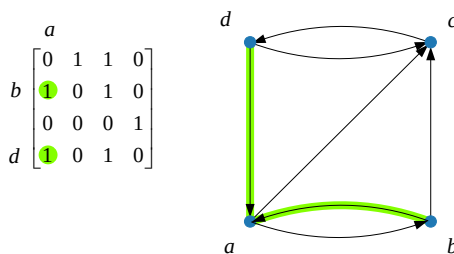


Figure 4. The Two Nodes That Directly Reach a — Column 1 of the Directed Graph's Adjacency Matrix $\mathbf{A}$

The number of paths from node a to node c passing through exactly one intermediate node can be obtained via the following matrix multiplication

$$A @ A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} @ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

As shown in Figure 5, the element in row 1, column 3 of the result of $A @ A$ is exactly the number of paths from node a to node c passing through exactly one intermediate node.

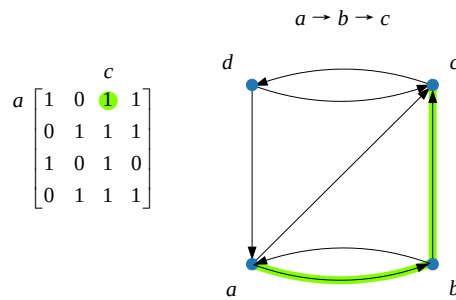


Figure 5. Paths From Node a to Node c Passing Through One Intermediate Node

As shown in Figure 6, the element in row 3, column 1 of the result of $A @ A$ is exactly the number of paths from node c to node a passing through exactly one intermediate node.

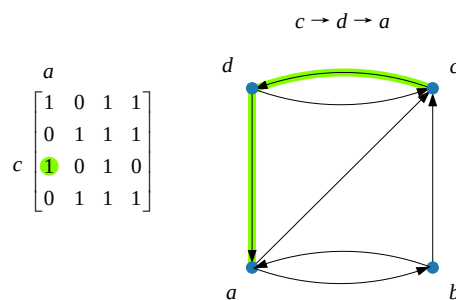


Figure 6. Paths From Node c to Node a Passing Through One Intermediate Node

? Please analyze, in the equation above, the meaning of the remaining elements in the result of $A @ A$.

To compute the number of paths from a to c that pass through at most one intermediate node, we can use the following operation

$$A + A @ A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 2 & 1 \end{bmatrix}$$

Please analyze the result above on your own.

Degree and Degree Matrix

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As with undirected graphs, the degree of any node in a directed graph is the number of edges connected to it.

However, since edges in a directed graph have direction, we're more interested in two concepts: in-degree and out-degree.

In a directed graph, a node's in-degree is the number of edges pointing into it — that is, the number of directed edges from other nodes to that node. Out-degree is the number of edges leaving it — that is, the number of directed edges from that node to other nodes. These two concepts describe a node's connectivity in a directed graph; the sum of in-degree and out-degree equals the node's degree.

Summing the directed graph's adjacency matrix $\mathbf{A}$ along its columns gives each node's in-degree, that is

$$\mathbf{1}^T \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 & 1 \end{bmatrix}$$

The in-degree matrix is a diagonal square matrix

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In addition, the directed graph's adjacency matrix $\mathbf{A}$'s Gram matrix $\mathbf{A}^T @ \mathbf{A}$'s main-diagonal elements are also the in-degree values

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}^T @ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Figure 7 shows the in-degrees of the directed graph's four nodes.

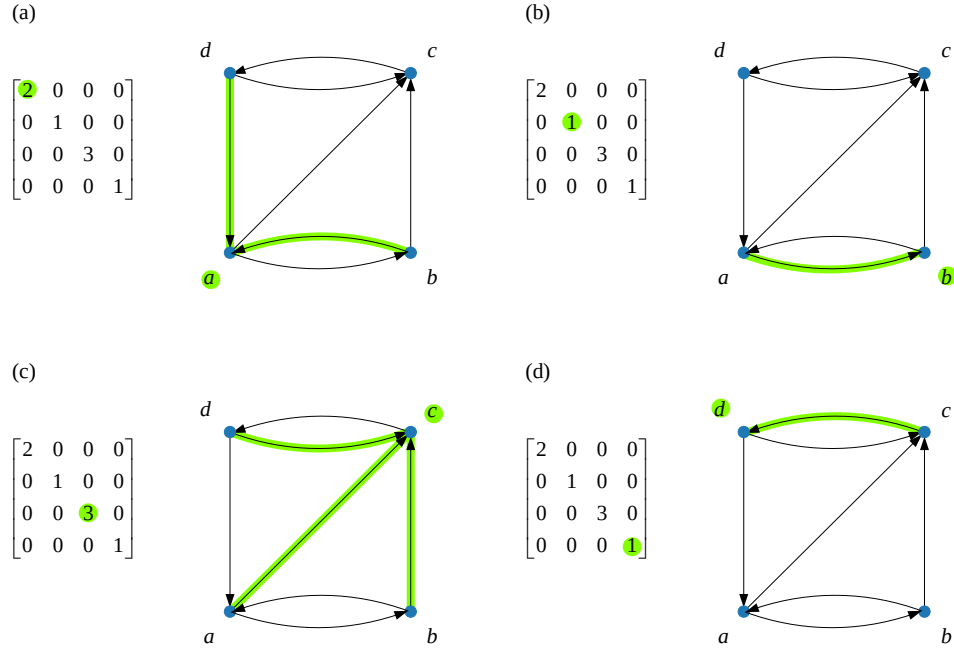


Figure 7. The In-Degrees of the Directed Graph's Four Nodes

Summing the directed graph's adjacency matrix A along its rows gives each node's out-degree, that is

$$A\mathbf{1} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 2 \end{bmatrix}$$

The out-degree matrix is also a diagonal square matrix

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

The main-diagonal elements of the directed graph's adjacency matrix A 's Gram matrix (of its transpose), $A @ A^T$, are also the out-degree values

$$AA^T = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} @ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 2 \end{bmatrix}$$

Figure 8 shows the out-degrees of the directed graph's four nodes.

⚠ Note that computing the out-degree and in-degree matrices via the Gram matrix only works for an unweighted directed graph.

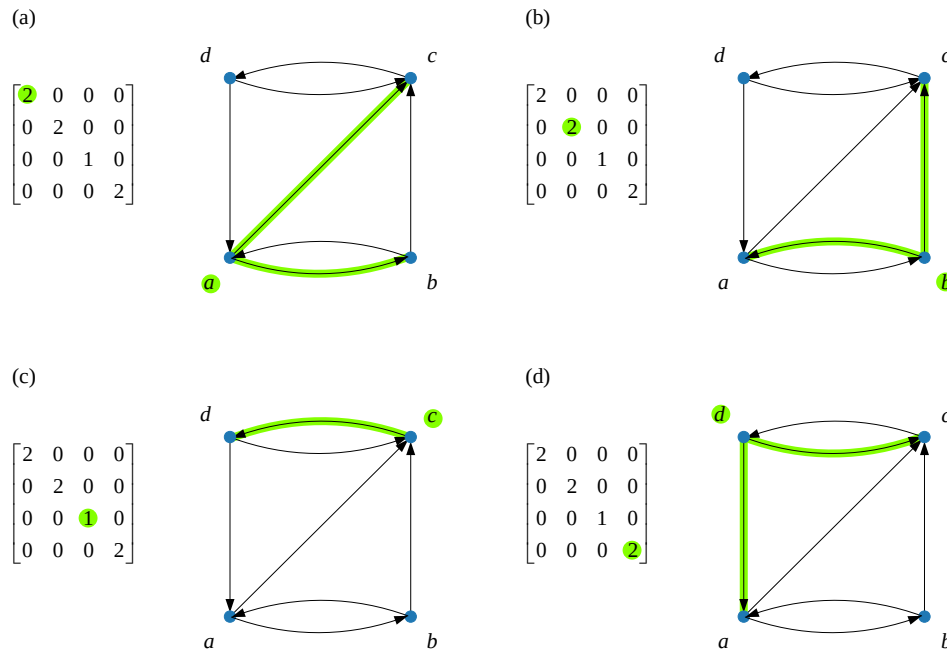


Figure 8. The Out-Degrees of the Directed Graph's Four Nodes

The Ball-Passing Problem

The adjacency matrix can be used to solve many interesting mathematical problems – the ball-passing problem, for example.



This ball-passing example is taken from The Way of Data.

Six students, a, b, c, d, e, and f, pass a ball among themselves. The rule is that each time someone passes the ball, they can pass it to anyone else, but not to themselves. Starting with a, after 4 passes the ball ends up back in a's hands – please compute how many ways this can happen.

Figure 9 shows one way this can happen, with the ball following the route $a \rightarrow f \rightarrow e \rightarrow b \rightarrow a$.

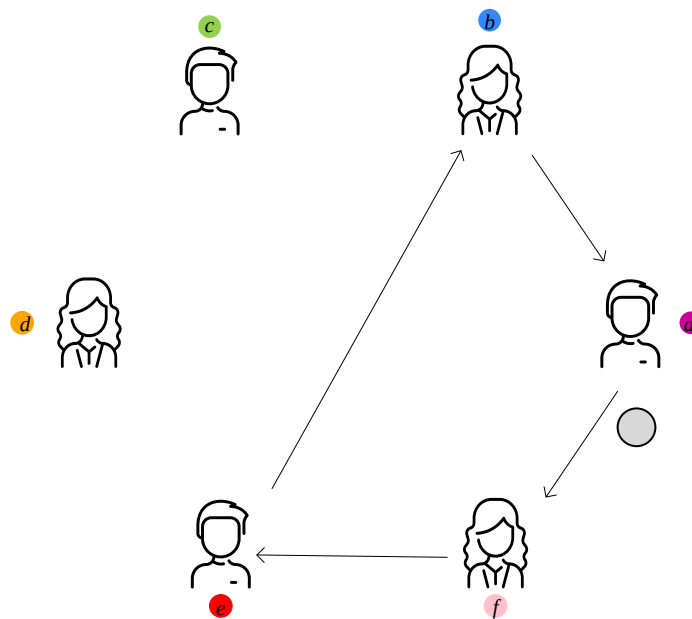


Figure 9. One Way the Ball Can Be Passed

Figure 10 shows all the possible routes for 4 passes; we need to find all the passing routes that start at *a* and return to *a*.

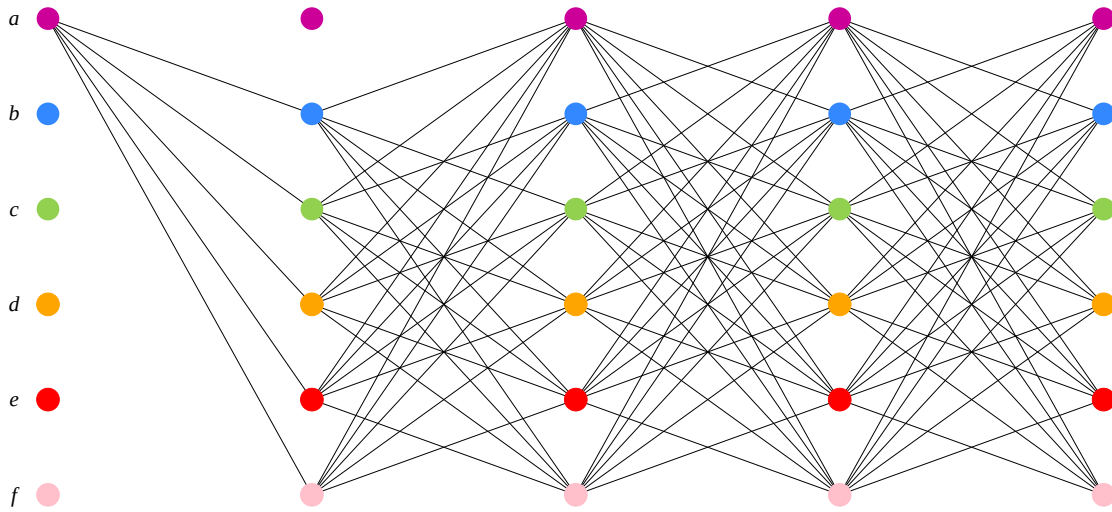


Figure 10. The Network of All Possible Routes

If we regard the six students *a*, *b*, *c*, *d*, *e*, and *f* as six nodes, the ball-passing relationships between them can be abstracted into the directed graph shown in Figure 11. This directed graph's adjacency matrix A is

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

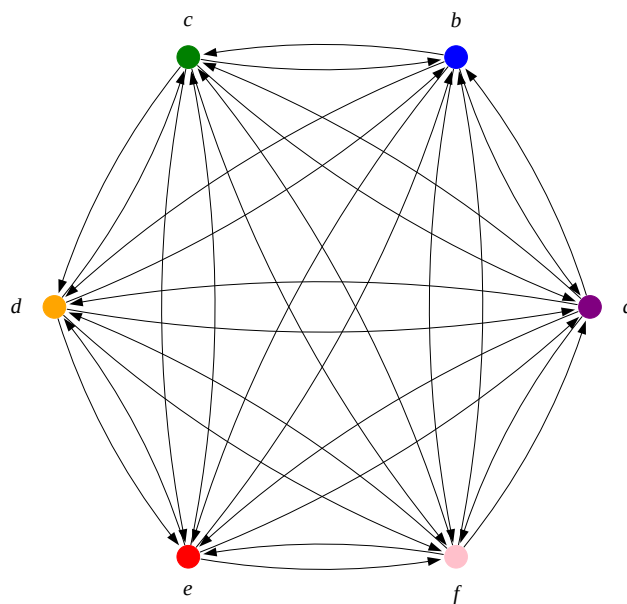


Figure 11. The Directed Graph Representing the Ball-Passing Problem

Let's talk about how to use adjacency matrix A to solve this ball-passing problem.

The First Pass

The ball starts in a's hands; we write this state as x_0

$$x_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The matrix multiplication Ax_0 represents the number of routes the ball, starting in a's hands, can take after the first pass. The specific result is

$$x_1 = Ax_0 = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

As shown in Figure 12, this result tells us that after one pass, the ball can be in the hands of any of the other five students besides a – that is, five possible routes. This is also the starting point for the second pass.

Summing all the elements of vector x_1 gives 5. This 5 actually represents 5^1 – as if, after one pass, "one becomes five."

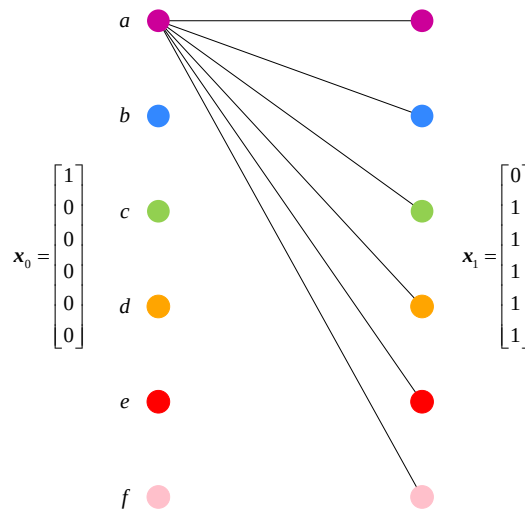
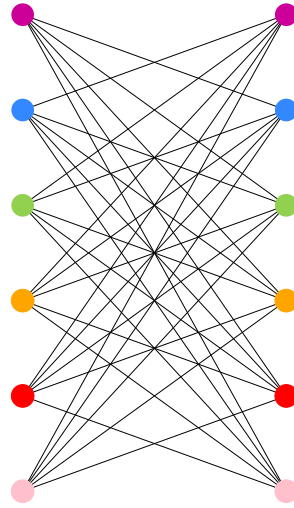


Figure 12. The Specific Meaning of the Matrix Multiplication $x_1 = Ax_0$

Matrix A 's elements sum to 30, that is, 6×5 (6 for the 6 nodes, and 5 for the 5 routes from each node). Figure 13 shows these 30 routes.

Figure 13. The Sum of All Elements of Matrix A

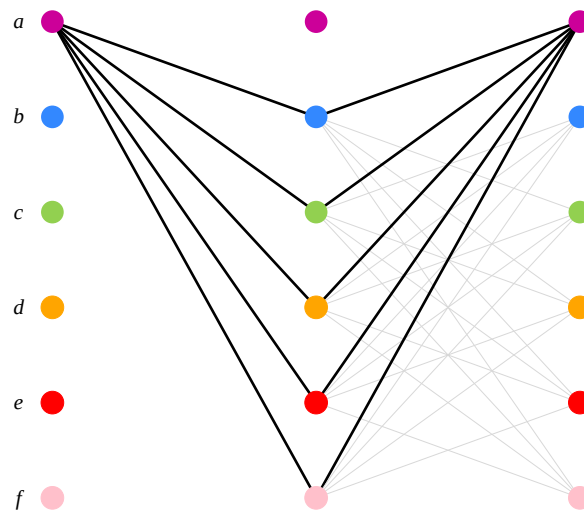
The Second Pass

As shown in Figure 14, the matrix multiplication Ax_1 represents the number of routes the ball, starting in a's hands, can take after the second pass. The specific result is

$$x_2 = Ax_1 = AAx_0 = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$$

For example, the 1st element of vector x_2 is 5, meaning there are 5 routes by which the ball returns to a's hands after 2 passes. Similarly, the 2nd element of vector x_2 is 4, meaning there are 4 routes by which the ball reaches b's hands after 2 passes.

Summing all the elements of vector x_2 gives 25, representing 5^2 — as if, after two passes, "one becomes five, and five becomes twenty-five."

Figure 14. The Specific Meaning of the Matrix Multiplication $x_2 = Ax_1$

Sharp-eyed readers may have already noticed that, in the equation above, the core operation is the power of square matrix A , namely A^2 . And A^2 's result is specifically

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$$A^2 = AA = \begin{bmatrix} 5 & 4 & 4 & 4 & 4 & 4 \\ 4 & 5 & 4 & 4 & 4 & 4 \\ 4 & 4 & 5 & 4 & 4 & 4 \\ 4 & 4 & 4 & 5 & 4 & 4 \\ 4 & 4 & 4 & 4 & 5 & 4 \\ 4 & 4 & 4 & 4 & 4 & 5 \end{bmatrix}$$

And, in the equation above, this is simply extracting the first column of A^2 's result.

From another angle, if we change this section's problem so that a different student starts with the ball, we only need to modify the initial state vector x_0

$$x_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

And for different initial state vectors x_0 , the operation $A^2 x_0$ extracts a different column of A^2 .

A^2 's result is also well worth a closer look!

A^2 's main-diagonal elements are all 5, representing the routes by which, after two passes, the ball returns to the same student who started with it.

Aside from the main-diagonal elements, all other elements of A^2 are 4. This result isn't surprising.

For example, if the ball starts in a's hands, there are 4 routes by which it ends up in b's hands after two passes. Since b can't pass to itself, that rules out one route. Also, a can't pass to itself and then pass to b, which rules out another route. This, in essence, is the core idea behind solving this problem using combinatorics.

And the sum of all of A^2 's elements is 150, that is, $6 \times 5 \times 5$.

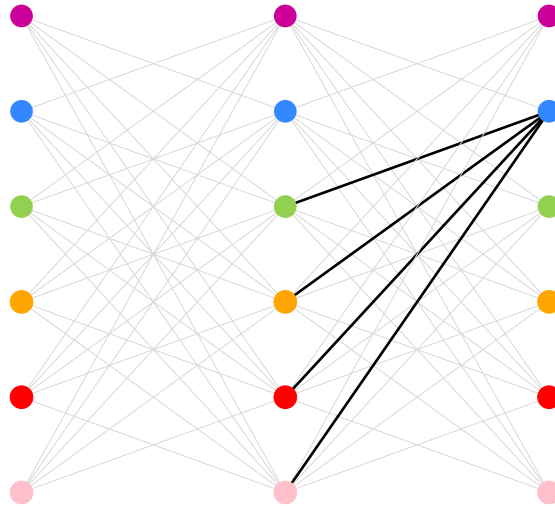
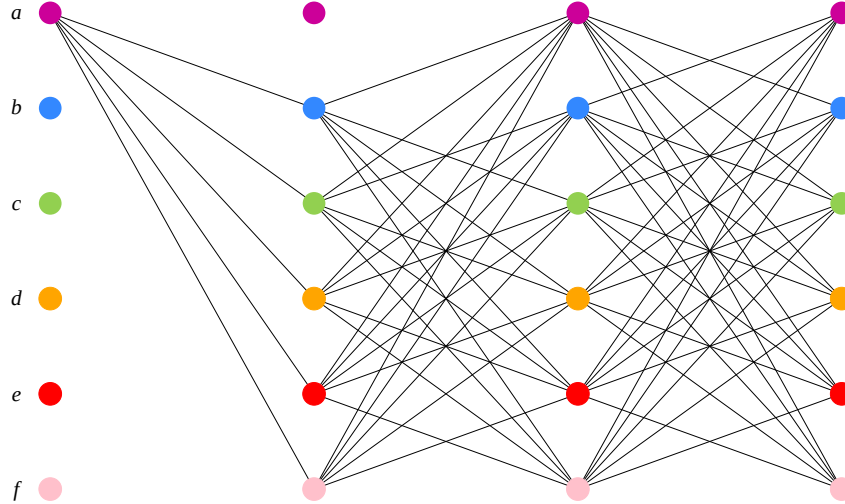


Figure 15. The Specific Meaning Represented by the Matrix Power A^2

The Third Pass

As shown in Figure 16, the matrix multiplication Ax_2 represents the number of routes the ball, starting in a's hands, can take after the third pass. The specific result is

$$\mathbf{x}_3 = \mathbf{A}\mathbf{x}_2 = \mathbf{A}\mathbf{A}\mathbf{A}\mathbf{x}_0 = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix} @ \begin{bmatrix} 5 \\ 4 \\ 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 20 \\ 21 \\ 21 \\ 21 \\ 21 \\ 21 \end{bmatrix}$$

Figure 16. The Specific Meaning of the Matrix Multiplication $\mathbf{x}_3 = \mathbf{A}\mathbf{x}_2$

And $\mathbf{A}^3$'s result is specifically

$$\mathbf{A}^3 = \mathbf{A}\mathbf{A}\mathbf{A} = \begin{bmatrix} 20 & 21 & 21 & 21 & 21 & 21 \\ 21 & 20 & 21 & 21 & 21 & 21 \\ 21 & 21 & 20 & 21 & 21 & 21 \\ 21 & 21 & 21 & 20 & 21 & 21 \\ 21 & 21 & 21 & 21 & 20 & 21 \\ 21 & 21 & 21 & 21 & 21 & 20 \end{bmatrix}$$

In the equation above, this is simply extracting the first column of the result in the equation above. Please analyze on your own why the main-diagonal elements of $\mathbf{A}^3$ are 20, while all other elements are 21.

The Fourth Pass

The matrix multiplication $\mathbf{A}\mathbf{x}_3$ represents the number of routes the ball, starting in a's hands, can take after the fourth pass. The specific result is

$$\mathbf{x}_4 = \mathbf{A}\mathbf{x}_3 = \mathbf{A}^4\mathbf{x}_0 = \begin{bmatrix} 105 \\ 104 \\ 104 \\ 104 \\ 104 \\ 104 \end{bmatrix}$$

The equation above tells us that the answer to the question posed at the start of this section is 105.

And $\mathbf{A}^4$'s result is specifically

$$A^4 = \begin{bmatrix} 105 & 104 & 104 & 104 & 104 & 104 \\ 104 & 105 & 104 & 104 & 104 & 104 \\ 104 & 104 & 105 & 104 & 104 & 104 \\ 104 & 104 & 104 & 105 & 104 & 104 \\ 104 & 104 & 104 & 104 & 105 & 104 \\ 104 & 104 & 104 & 104 & 104 & 105 \end{bmatrix}$$

? Please think about this: if the ball is passed no more than four times, starting with a, how many ways are there for it to end up in f's hands?



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please use NetworkX to construct and draw each of the directed graphs in Table 1 one by one.

Q2. Please compute the adjacency matrices of all the directed graphs in Table 1.

Q3. Please compute the out-degrees and in-degrees of the nodes for each directed graph in Table 1.

Q4. Four students pass a ball among themselves. The rule is still that each time someone passes the ball, they can pass it to anyone else, but not to themselves. Starting with a, if the ball is passed no more than 3 times and ends up back in a's hands, please compute how many ways this can happen.

Q5. Please use NetworkX to draw the directed graph shown in Figure 11, and compute its adjacency matrix.